A RANK $(2g-1)$ AFFINE-PRYM CONSTRUCTION AND ITS TWO-BLOCK OPTIMALITY

[AUTHOR NAME]

Abstract. Let Σ_g be a closed connected oriented surface of genus $g \geq 2$, and let

$$\psi : \pi_1(\Sigma_g) \longrightarrow \mathbb{C}^\times$$

be a non-trivial finite scalar character. Put

$$V_\psi := H^1(\Sigma_g; \mathbb{C}_\psi), \qquad \dim_{\mathbb{C}} V_\psi = 2g-2.$$

We construct an explicit rank $2g-1$ non-semisimple representation of $\pi_1(\Sigma_g)$ with infinite image and finite ordinary mapping-class-group orbit. The construction is the universal affine-Prym extension attached to the full Prym cohomology space V_ψ. For punctured surfaces, the same construction applies to $W_1 H^1(\Sigma_{g,n}; \mathbb{C}_\psi) \cong H^1(\Sigma_g; \mathbb{C}_\psi)$ under peripheral triviality.

The second result proves that this rank is optimal inside the restricted class of scalar two-block triangular Prym extensions

$$0 \longrightarrow \mathbb{C}_\psi \otimes M \longrightarrow \rho \longrightarrow \mathbb{C} \otimes N \longrightarrow 0.$$

Namely, if such an extension has infinite image and finite ordinary mapping-class-group orbit, then

$$\dim M + \dim N \geq \dim V_\psi + 1 = 2g-1.$$

The proof uses two external inputs: Looijenga's cyclic Prym image theorem and Westwick's bound for complex linear spaces of matrices of fixed rank. The Looijenga input is used only through its single-character projective consequence: the connected projective Zariski closure of the stabilizer of ψ acts on $\mathbb{P}(V_\psi^\vee)$ as $\mathrm{PSp}(V_\psi)$ when $\psi^2 = 1$ and as $\mathrm{PSL}(V_\psi)$ when $\psi^2 \neq 1$.

Contents

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1. Introduction

Let

$$\Gamma_g = \pi_1(\Sigma_g), \qquad g \geq 2,$$

where Σ_g is a closed connected oriented surface. A finite character

$$\psi : \Gamma_g \longrightarrow \mathbb{C}^\times$$

factors through $H_1(\Sigma_g; \mathbb{Z})$ and hence through a finite cyclic quotient. Let $\mathbb{C}_\psi$ be the corresponding rank-one local system. If $\psi \neq 1$, then

$$H^0(\Sigma_g; \mathbb{C}_\psi) = 0, \qquad H^2(\Sigma_g; \mathbb{C}_\psi) = 0,$$

the second equality following from Poincaré duality and the vanishing of $H^0(\Sigma_g; \mathbb{C}_{\psi^{-1}})$. Therefore

$$\dim_{\mathbb{C}} H^1(\Sigma_g; \mathbb{C}_\psi) = 2g - 2. \tag{1}$$

We write

$$V_\psi := H^1(\Sigma_g; \mathbb{C}_\psi).$$

The first point of this paper is that the full Prym cohomology space gives a rank $2g-1$ affine-unipotent example. More precisely, the identity tensor

$$\mathrm{id}_{V_\psi} \in V_\psi \otimes V_\psi^\vee$$

defines a triangular representation of rank

$$1 + \dim V_\psi = 2g - 1$$

with infinite image and finite ordinary mapping-class-group orbit. This is the existence, or upper-bound, part of the paper.

The second point is a restricted optimality statement. We consider only scalar two-block triangular Prym extensions: representations whose semisimplification has exactly two scalar isotypical types, copies of $\mathbb{C}_\psi$ and copies of the trivial character. In this restricted category, the rank $2g-1$ construction cannot be compressed.

Main results.

Theorem 1.1 (Rank $2g-1$ affine-Prym examples)**.** *Let $g \geq 2$ and let $\psi : \pi_1(\Sigma_g) \to \mathbb{C}^\times$ be a non-trivial finite character. Set $V_\psi = H^1(\Sigma_g; \mathbb{C}_\psi)$. There exists a representation*

$$\rho_\psi^{\mathrm{univ}} : \pi_1(\Sigma_g) \longrightarrow \mathrm{GL}_{2g-1}(\mathbb{C})$$

of the form

$$\rho_\psi^{\mathrm{univ}}(\gamma) = \begin{pmatrix} \psi(\gamma) I_{V_\psi^\vee} & U(\gamma) \\ 0 & 1 \end{pmatrix},$$

where U is a cocycle representing the identity tensor in $V_\psi \otimes V_\psi^\vee$, such that:

(a) *$\rho_\psi^{\mathrm{univ}}$ has rank $1 + \dim V_\psi = 2g-1$;*
(b) *its image is infinite;*
(c) *its ordinary $\mathrm{Mod}(\Sigma_g)$-orbit is finite.*

The same construction applies to a punctured surface $\Sigma_{g,n}$, under Convention 2.6, with V_ψ replaced by $W_1 H^1(\Sigma_{g,n}; \mathbb{C}_\psi) \cong H^1(\Sigma_g; \mathbb{C}_\psi)$.

Theorem 1.2 (Restricted scalar two-block optimality)**.** *Let $g \geq 2$, let $\psi : \pi_1(\Sigma_g) \to \mathbb{C}^\times$ be a non-trivial finite character, and put $V_\psi = H^1(\Sigma_g; \mathbb{C}_\psi)$. Let*

$$0 \longrightarrow \mathbb{C}_\psi \otimes M \longrightarrow \rho \longrightarrow \mathbb{C} \otimes N \longrightarrow 0$$

be a scalar two-block triangular Prym extension with $M, N \neq 0$. Assume that:

(i) *the ordinary $\mathrm{Mod}(\Sigma_g)$-orbit of the conjugacy class of ρ is finite;*
(ii) *the image of ρ is infinite.*

Then

$$\dim M + \dim N \geq \dim V_\psi + 1 = 2g-1.$$

Consequently, the rank $2g-1$ affine-Prym construction of Theorem 1.1 is optimal inside the scalar two-block triangular Prym category.

Warning 1.3 (Claims not made)**.** *This paper proves a restricted optimality theorem, not a global minimal-rank theorem. It does not rule out smaller-rank examples outside the scalar two-block triangular Prym framework. In particular, it makes no claim about:*

- *non-scalar finite-image semisimple blocks;*
- *representations with longer non-semisimple filtrations;*
- *semisimple finite-orbit representations;*
- *point-pushing, quantum, TQFT, or other non-Prym constructions;*
- *the global minimal rank among all finite-orbit representations.*

Thus any example of rank $< 2g-1$, if it exists, must use a mechanism outside the scalar two-block triangular Prym framework studied here.

2. The affine-Prym construction

This section proves Theorem 1.1. We begin with the closed case and then record the punctured version under peripheral triviality.

2.1. **The universal extension class.** Let

$$V_\psi = H^1(\Sigma_g; \mathbb{C}_\psi), \qquad M := V_\psi^\vee, \qquad N := \mathbb{C}.$$

Then

$$\operatorname{Ext}^1_{\Gamma_g}(\mathbb{C}, \mathbb{C}_\psi \otimes M) \cong H^1(\Sigma_g; \mathbb{C}_\psi) \otimes M = V_\psi \otimes V_\psi^\vee = \operatorname{End}(V_\psi).$$

Let

$$x_{\mathrm{univ}} := \mathrm{id}_{V_\psi} \in V_\psi \otimes V_\psi^\vee.$$

Choose a cocycle

$$U \in Z^1(\Gamma_g; \mathbb{C}_\psi \otimes V_\psi^\vee)$$

representing x_{univ}. Thus

$$U(\gamma\delta) = U(\gamma) + \psi(\gamma) U(\delta).$$

Define

$$\rho_\psi^{\mathrm{univ}}(\gamma) = \begin{pmatrix} \psi(\gamma) I_{V_\psi^\vee} & U(\gamma) \\ 0 & 1 \end{pmatrix}.$$

Here $U(\gamma)$ is viewed as a linear map $\mathbb{C} \to V_\psi^\vee$.

Lemma 2.1 (Representation identity)**.** *The assignment $\gamma \mapsto \rho_\psi^{\mathrm{univ}}(\gamma)$ is a representation of Γ_g.*

Proof. Multiplying the two matrices gives

$$\begin{pmatrix} \psi(\gamma) I & U(\gamma) \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \psi(\delta) I & U(\delta) \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \psi(\gamma\delta) I & U(\gamma) + \psi(\gamma) U(\delta) \\ 0 & 1 \end{pmatrix}.$$

The upper-right entry is $U(\gamma\delta)$ by the cocycle identity. □

2.2. **Infinite image.**

Lemma 2.2 (Non-vanishing on the kernel)**.** *Let $K = \ker \psi$. For any cocycle U representing the non-zero class x_{univ}, the restriction $U|_K$ is not identically zero.*

Proof. Suppose $U|_K = 0$. Since K acts trivially on the coefficient module $\mathbb{C}_\psi \otimes V_\psi^\vee$, the cocycle U descends to a cocycle of the finite group $\Gamma_g/K = \operatorname{im}(\psi)$. Over characteristic zero, the cohomology of a finite group with coefficients in a finite-dimensional complex representation vanishes in positive degree. Hence the descended cocycle is a coboundary. Pulling back, U is a coboundary on Γ_g, contradicting the fact that $x_{\mathrm{univ}} \neq 0$ in $H^1(\Sigma_g; \mathbb{C}_\psi) \otimes V_\psi^\vee$. □

Corollary 2.3 (Infinite image)**.** *The image of $\rho_\psi^{\mathrm{univ}}$ is infinite.*

Proof. By Lemma 2.2, choose $\gamma \in K$ such that $U(\gamma) \neq 0$. Then $\psi(\gamma) = 1$ and

$$\rho_\psi^{\mathrm{univ}}(\gamma) = \begin{pmatrix} I & U(\gamma) \\ 0 & 1 \end{pmatrix}$$

is a non-trivial unipotent element. Its powers have upper-right entry $nU(\gamma)$, so they are pairwise distinct. □

2.3. **Finite ordinary mapping-class-group orbit.** We use ordinary conjugacy throughout: no projectivization, twisting equivalence, or GIT quotient is imposed.

Lemma 2.4 (Stabilizer fixes the universal class up to ordinary conjugacy)**.** *Let*

$$H_\psi = \mathrm{Stab}_{\mathrm{Mod}(\Sigma_g)}(\psi).$$

Then the ordinary conjugacy class of $\rho_\psi^{\mathrm{univ}}$ is fixed by H_ψ.

Proof. For $h \in H_\psi$, the transformed class lies again in $V_\psi \otimes V_\psi^\vee$. Under the identification

$$V_\psi \otimes V_\psi^\vee \cong \mathrm{End}(V_\psi),$$

the universal class is the identity. The action of h on the first tensor factor transforms the associated contraction map by the dual action on $V_\psi^\vee$. Changing the basis of the multiplicity space $M = V_\psi^\vee$ by the inverse dual automorphism sends the transformed contraction map back to the identity. Thus the transformed extension class is in the same $\mathrm{GL}(V_\psi^\vee)$-orbit, hence gives the same ordinary conjugacy class. □

Lemma 2.5 (Finite character orbit)**.** *The* $\mathrm{Mod}(\Sigma_g)$*-orbit of a finite character ψ is finite.*

Proof. If the order of ψ divides q, then ψ factors through the finite group $H_1(\Sigma_g; \mathbb{Z}/q\mathbb{Z})$. There are only finitely many characters of this finite group. □

Proof of Theorem 1.1 in the closed case. The rank is $1 + \dim V_\psi = 2g - 1$. Infinite image follows from Corollary 2.3. By Lemma 2.5, the character orbit is finite. For each character in that finite orbit, its stabilizer fixes the corresponding universal ordinary conjugacy class by Lemma 2.4. Therefore the full ordinary mapping-class-group orbit of $\rho_\psi^{\mathrm{univ}}$ is finite. □

2.4. **Punctured surfaces under peripheral triviality.** Let $\Sigma_{g,n}$ be a punctured surface, and let $\bar{\Sigma}_{g,n}$ denote the closed surface obtained by filling the punctures. Thus $\bar{\Sigma}_{g,n} \cong \Sigma_g$.

Convention 2.6 (Peripheral triviality)**.** *For a punctured surface $\Sigma_{g,n}$, assume that ψ is trivial on peripheral loops and that the triangular representation has trivial peripheral monodromy. In matrix form, this means that the cocycle U vanishes on each puncture loop.*

Under Convention 2.6, the relevant cohomology is the weight-one part
$$W_1 H^1(\Sigma_{g,n}; \mathbb{C}_\psi) \cong H^1(\Sigma_g; \mathbb{C}_\psi).$$
Applying the preceding construction to this weight-one Prym block gives the same rank
$$1 + \dim W_1 H^1(\Sigma_{g,n}; \mathbb{C}_\psi) = 2g - 1.$$
The peripheral loops are sent to the identity because ψ is peripheral-trivial and the cocycle is required to vanish on them. Thus the construction descends to the filled Prym block and has the same finite-orbit and infinite-image properties. Without peripheral triviality, additional boundary cohomology classes appear, and this paper makes no statement in that generality.

3. Looijenga input

The proof of the lower bound uses only one deep Prym monodromy input: Looijenga's cyclic image theorem. It does not use Looijenga's multi-factor independence theorem.

3.1. **The cyclic quotient attached to a character.** Let $\psi : \pi_1(\Sigma_g) \to \mathbb{C}^\times$ be finite and non-trivial. It factors through $H_1(\Sigma_g; \mathbb{Z})$. Let
$$q : H_1(\Sigma_g; \mathbb{Z}) \twoheadrightarrow C$$
be the finite cyclic quotient determined by ψ, so that pullback identifies
$$q^*(\widehat{C}) = \langle \psi \rangle \subset \mathrm{Hom}(H_1(\Sigma_g; \mathbb{Z}), \mathbb{C}^\times), \qquad \widehat{C} := \mathrm{Hom}(C, \mathbb{C}^\times).$$
Equivalently, C is the cyclic deck group of the connected cyclic cover that realizes the ψ-Prym eigenspace.

Let
$$H_\psi := \mathrm{Stab}_{\mathrm{Mod}(\Sigma_g)}(\psi).$$
Looijenga's subgroup associated with C, denoted here by S_C, is the subgroup of mapping classes inducing the identity on the quotient C. This is equivalently the subgroup fixing $\widehat{C}$ pointwise under the dual pullback action. Since $q^*(\widehat{C}) = \langle \psi \rangle$ is generated by ψ, fixing ψ is equivalent to fixing every power of ψ, hence to fixing $q^*(\widehat{C})$ pointwise. Thus, in this single-character setup,
$$S_C = H_\psi.$$
The point here is the identity action on the quotient C, not merely preservation of the kernel of q or preservation of the corresponding cover.

3.2. **Looijenga's cyclic image theorem.** If $S_C^{\#}$ denotes the group of lifts of mapping classes inducing the identity on the cyclic quotient C, then there is a central extension
$$1 \longrightarrow C \longrightarrow S_C^{\#} \longrightarrow S_C \longrightarrow 1,$$
where the kernel is the deck group. Here $S_C^{\#}$ is our abbreviation for Looijenga's lifted group $\mathcal{S}^{\#}_{;C}$.

External Theorem 3.1 (Looijenga's cyclic Prym image theorem)**.** *Let C be a non-trivial finite cyclic quotient of $H_1(\Sigma_g;\mathbb{Z})$, let $\widetilde{\Sigma}_C \to \Sigma_g$ be the associated connected cyclic cover, and let H_C be Looijenga's cyclic Prym module over the cyclotomic integer ring R_C. Then H_C is a free R_C-module of rank $2(g-1)$ with its standard skew-Hermitian intersection form. Moreover, the image of the lifted mapping class group $S_C^{\#}$ on H_C is the arithmetic group denoted by $U^{\#}(H_C)$.*

All notation in External Theorem 3.1 follows Looijenga, Section 2. The freeness and form statement are Looijenga's Proposition 2.2, and the image statement is the cyclic case of Looijenga's Theorem 2.4.

Remark 3.2 (No multi-factor theorem)*.* The paper uses only the one cyclic factor C attached to $\langle\psi\rangle$. No simultaneous-independence theorem for several cyclic factors is invoked.

3.3. **Deck transformations disappear projectively.** Let

$$E_\psi := H_C \otimes_{R_C,\psi} \mathbb{C}$$

be the complex ψ-eigenspace after scalar extension through the embedding determined by ψ. A deck transformation $c \in C$ acts on E_ψ by the scalar $\psi(c)$, or by $\psi(c)^{-1}$, depending on the left/right module convention. In either convention the action is scalar. Hence the deck kernel acts trivially on $\mathbb{P}(E_\psi)$.

Thus the projective image of the lifted group $S_C^{\#}$ on $\mathbb{P}(E_\psi)$ and the projective image of the ordinary stabilizer $S_C = H_\psi$ have the same identity component of their projective Zariski closures.

3.4. **Homology versus cohomology.** With the convention used here, the ψ-eigenspace of the homology of the cyclic cover identifies with twisted homology with inverse character:

$$E_\psi \cong H_1(\Sigma_g;\mathbb{C}_{\psi^{-1}}).$$

By Poincaré duality and the universal coefficient theorem over $\mathbb{C}$,

$$E_\psi^\vee \cong H^1(\Sigma_g;\mathbb{C}_\psi) = V_\psi.$$

Thus $E_\psi \cong V_\psi^\vee$ after fixing the standard dual convention. If the opposite left/right convention is used, replace ψ by ψ^{-1}; this exchanges the two conjugate eigenspaces and does not affect the final projective closure conclusion, since dualizing preserves PSp / PSL and the dichotomy $\psi^2 = 1$ versus $\psi^2 \neq 1$.

3.5. **The single-character projective consequence.** The next lemma is the passage from Looijenga's K_C/K_C^0-unitary group to one complex character block. It is not an additional mapping-class-group input; the only mapping-class-group input remains External Theorem 3.1.

Lemma 3.3 (Classical group on one character block)**.** *Assume the image statement of External Theorem 3.1. Let*

$$E_\psi = H_C \otimes_{R_C,\psi} \mathbb{C}$$

be the complex eigenspace attached to a non-trivial character $\psi \in \widehat{C}$. Then the identity component of the projective Zariski closure of $U^\#(H_C)$ on $\mathbb{P}(E_\psi)$ is

$$\begin{cases} \mathrm{PSp}(E_\psi), & \psi^2 = 1, \\ \mathrm{PSL}(E_\psi), & \psi^2 \neq 1. \end{cases}$$

Proof. Write $K = K_C$ and $K^+ = K_C^0$. Looijenga's form on H_C is a non-degenerate skew-Hermitian form over R_C. The corresponding K^+-algebraic unitary group has integral arithmetic subgroup $U(H_C)$, and Looijenga's group $U^\#(H_C)$ is obtained by imposing that the R_C-determinant be a square of a root of unity. Equivalently, it differs from the special unitary arithmetic group $\mathrm{SU}(H_C)$ only by finite determinant/root-of-unity cosets. These finite cosets do not change the identity component of the complex projective Zariski closure. Under Looijenga's identification of the mapping-class image with $U^\#(H_C)$, the connected complex Zariski closure is therefore obtained by base-changing the corresponding K^+-algebraic special unitary group.

First suppose $\psi^2 = 1$. Since $\psi \neq 1$, the character ψ has order two. The block E_ψ is self-dual. In this case $K = K^+ = \mathbb{Q}$, the involution is trivial, and Looijenga's skew-Hermitian form becomes, after scalar extension to $\mathbb{C}$, a non-degenerate alternating complex bilinear form on E_ψ. Thus the connected complex algebraic group on the block is the symplectic group $\mathrm{Sp}(E_\psi)$, not an orthogonal group: the form is skew-symmetric, equivalently alternating, rather than symmetric. Therefore the induced connected projective group is $\mathrm{PSp}(E_\psi)$.

Now suppose $\psi^2 \neq 1$. Then ψ and ψ^{-1} are distinct conjugate characters. After base change through the real embedding of K^+ determined by $\psi|_{K^+}$, one has

$$K \otimes_{K^+} \mathbb{C} \cong \mathbb{C} \oplus \mathbb{C},$$

with the two factors corresponding to ψ and ψ^{-1}. Hence

$$H_C \otimes_{K^+} \mathbb{C} = E_\psi \oplus E_{\psi^{-1}}.$$

The skew-Hermitian form pairs E_ψ perfectly with $E_{\psi^{-1}}$, and each of these two summands is isotropic. The complexified special unitary group therefore acts as

$$A \oplus (A^{-1})^\vee, \qquad A \in \mathrm{SL}(E_\psi).$$

Consequently the induced connected projective group on $\mathbb{P}(E_\psi)$ is $\mathrm{PSL}(E_\psi)$. The determinant, norm-one, and root-of-unity restrictions built into $U^\#(H_C)$ only add finite components or finite cosets, and hence do not change the identity component of the complex projective Zariski closure. $\square$

Proposition 3.4 (Single-character projective consequence)**.** *Let $\psi : \pi_1(\Sigma_g) \to \mathbb{C}^\times$ be a non-trivial finite character, and set*

$$V_\psi = H^1(\Sigma_g; \mathbb{C}_\psi), \qquad H_\psi = \mathrm{Stab}_{\mathrm{Mod}(\Sigma_g)}(\psi).$$

Let

$$\rho_\psi^{\mathrm{proj}} : H_\psi \longrightarrow \mathrm{PGL}(V_\psi^\vee)$$

be the induced projective representation. Then

$$\left(\overline{\rho_\psi^{\mathrm{proj}}(H_\psi)}^{\mathrm{Zar}}\right)^0 = \begin{cases} \mathrm{PSp}(V_\psi), & \psi^2 = 1, \\ \mathrm{PSL}(V_\psi), & \psi^2 \neq 1, \end{cases}$$

where the groups on the right are viewed through the natural dual projective action on $\mathbb{P}(V_\psi^\vee)$. If $H_0 \leq H_\psi$ is a finite-index subgroup, then

$$\left(\overline{\rho_\psi^{\mathrm{proj}}(H_0)}^{\mathrm{Zar}}\right)^0 = \left(\overline{\rho_\psi^{\mathrm{proj}}(H_\psi)}^{\mathrm{Zar}}\right)^0.$$

Proof. Choose the cyclic quotient $q : H_1(\Sigma_g; \mathbb{Z}) \twoheadrightarrow C$ with $q^*(\widehat{C}) = \langle \psi \rangle$. By External Theorem 3.1, the lifted group $S_C^\#$ has image $U^\#(H_C)$ on Looijenga's cyclic Prym module. The equality $S_C = H_\psi$ follows from the choice of C, as explained in Section 3.1. The deck kernel acts by scalars on the ψ-eigenspace and hence disappears in projective space. The eigenspace is $E_\psi \cong V_\psi^\vee$, with the harmless ψ versus ψ^{-1} convention described in Section 3.4. Now Lemma 3.3 gives the asserted projective Zariski closure.

For the finite-index assertion, let $H_0 \leq H_\psi$ be a finite-index subgroup. Then $\rho(H_\psi)$ is a finite union of cosets of $\rho(H_0)$. Therefore the Zariski closure of $\rho(H_\psi)$ is a finite union of translates of the Zariski closure of $\rho(H_0)$, and a finite union of translates does not alter the connected component of the identity. ☐

Corollary 3.5 (Finite-orbit projective subvarieties are trivial)**.** *Let $Z \subset \mathbb{P}(V_\psi^\vee)$ be a closed subvariety. If the H_ψ-orbit of Z is finite, then*

$$Z = \varnothing \qquad \text{or} \qquad Z = \mathbb{P}(V_\psi^\vee).$$

The same conclusion holds with H_ψ replaced by any finite-index subgroup.

Proof. If the orbit of Z is finite, some finite-index subgroup $H_0 \leq H_\psi$ stabilizes Z. The stabilizer of a closed projective subvariety is Zariski closed in the relevant projective linear group. By Proposition 3.4, the connected projective Zariski closure of H_0 is either $\mathrm{PSp}(V_\psi)$ or $\mathrm{PSL}(V_\psi)$. Both groups act transitively on $\mathbb{P}(V_\psi^\vee)$. For PSL this is standard; for PSp, every nonzero vector in a symplectic vector space is isotropic and can be included in a symplectic basis, so any two lines can be carried to one another by a symplectic automorphism. Hence Z is either empty or the whole projective space. ☐

4. Extension classes and ordinary conjugacy

Fix non-zero finite-dimensional complex vector spaces M and N, and put

$$m = \dim M, \qquad n = \dim N, \qquad U = \mathrm{Hom}(N, M).$$

A scalar two-block triangular Prym extension is an extension

$$0 \longrightarrow \mathbb{C}_\psi \otimes M \longrightarrow \rho \longrightarrow \mathbb{C} \otimes N \longrightarrow 0.$$

After choosing a vector-space splitting, it has the form

$$\rho(\gamma) = \begin{pmatrix} \psi(\gamma) I_M & u(\gamma) \\ 0 & I_N \end{pmatrix},$$

where u is a cocycle with values in $\mathbb{C}_\psi \otimes \mathrm{Hom}(N, M)$. The extension class lies in

$$\mathrm{Ext}^1_{\Gamma_g}(\mathbb{C} \otimes N, \mathbb{C}_\psi \otimes M) \cong H^1(\Sigma_g; \mathbb{C}_\psi) \otimes \mathrm{Hom}(N, M) = V_\psi \otimes U.$$

Write this class as

$$x \in V_\psi \otimes U.$$

Changing bases in M and N gives the group

$$K = \mathrm{GL}(M) \times \mathrm{GL}(N),$$

acting on $U = \mathrm{Hom}(N, M)$ by

$$(A, B) \cdot T = ATB^{-1}.$$

It acts on $V_\psi \otimes U$ through the second tensor factor. The notation $(V_\psi \otimes U)/K$ denotes the set of ordinary K-orbits, not a GIT quotient.

Lemma 4.1 (Infinite image and non-zero extension class)**.** *For a scalar two-block triangular Prym extension, the image of ρ is finite if and only if its extension class x is zero. In particular, infinite image implies $x \neq 0$.*

Proof. If $x = 0$, the extension splits as $(\mathbb{C}_\psi \otimes M) \oplus (\mathbb{C} \otimes N)$. Both diagonal characters have finite image, so the image of ρ is finite.

Conversely, if the image of ρ is finite, then ρ factors through a finite group. Over $\mathbb{C}$, every representation of a finite group is semisimple by Maschke's theorem. Hence the extension splits, and its extension class is zero. ☐

Lemma 4.2 (Ordinary conjugacy and K-orbits)**.** *Fix the semisimplification*

$$(\mathbb{C}_\psi \otimes M) \oplus (\mathbb{C} \otimes N).$$

Two scalar two-block extensions with classes $x, x' \in V_\psi \otimes \mathrm{Hom}(N, M)$ are isomorphic as Γ_g-local systems if and only if x' lies in the $K = \mathrm{GL}(M) \times \mathrm{GL}(N)$-orbit of x.

Proof. Let $A = \mathbb{C}_\psi \otimes M$ and $B = \mathbb{C} \otimes N$. Since $\psi \neq 1$,

$$\mathrm{Hom}_{\Gamma_g}(B, A) = 0, \qquad \mathrm{Hom}_{\Gamma_g}(A, B) = 0.$$

The simple rank-one local systems $\mathbb{C}_\psi$ and $\mathbb{C}$ are non-isomorphic, so every isomorphism between two extensions with this fixed semisimplification preserves the two scalar isotypical constituents on the associated semisimple quotient. It induces automorphisms of A and B, hence elements of $\mathrm{GL}(M)$ and $\mathrm{GL}(N)$.

The natural action of

$$\mathrm{Aut}_{\Gamma_g}(A) \times \mathrm{Aut}_{\Gamma_g}(B) = \mathrm{GL}(M) \times \mathrm{GL}(N)$$

on $\mathrm{Ext}^1_{\Gamma_g}(B, A)$ is exactly the action on the $\mathrm{Hom}(N, M)$ factor. Changing the vector-space splitting changes the cocycle representative by a coboundary and hence not the cohomology class. Thus ordinary isomorphism classes with fixed semisimplification are precisely the K-orbits. □

Lemma 4.3 (Finite ordinary orbit gives finite K-orbit data)**.** *If the ordinary* $\mathrm{Mod}(\Sigma_g)$*-orbit of the conjugacy class of* ρ *is finite, then the* K*-orbit*

$$[x] \in (V_\psi \otimes U)/K$$

has finite orbit under H_ψ*.*

Proof. Restrict the finite $\mathrm{Mod}(\Sigma_g)$-orbit to H_ψ. For every $h \in H_\psi$, the transformed representation has the same scalar semisimplification, since h fixes ψ and fixes the trivial character. Only finitely many ordinary conjugacy classes occur. By Lemma 4.2, each such conjugacy class with this fixed semisimplification determines one K-orbit of extension classes. Hence only finitely many K-orbits occur. □

5. Full Prym support

Definition 5.1 (Prym support)*.* For $x \in V_\psi \otimes U$, define its support in the Prym factor by

$$S(x) := \mathrm{span}\{(\mathrm{id}_{V_\psi} \otimes \lambda)(x) : \lambda \in U^\vee\} \subset V_\psi.$$

Equivalently, $S(x)$ is the smallest linear subspace $S \subset V_\psi$ such that $x \in S \otimes U$.

Lemma 5.2 (Equivariance of support)**.** *For* $k \in K$ *and* $h \in H_\psi$*,*

$$S(kx) = S(x), \qquad S(hx) = hS(x).$$

Proof. The group K acts invertibly on the U-factor and trivially on the V_ψ-factor, so contraction by all elements of $U^\vee$ spans the same subspace. The mapping class group acts on the extension class through the Prym factor, so contraction in U commutes with the induced action on V_ψ. □

Lemma 5.3 (No finite-index invariant proper subspace)**.** *No finite-index subgroup of* H_ψ *fixes a non-zero proper linear subspace of* V_ψ*.*

Proof. Suppose $0 \neq W \subsetneq V_\psi$ is fixed by a finite-index subgroup $H_0 \leq H_\psi$. Its annihilator $W^\perp \subset V_\psi^\vee$ is non-zero and proper. Therefore $\mathbb{P}(W^\perp)$ is a non-empty proper closed subvariety of $\mathbb{P}(V_\psi^\vee)$. Since H_0 is finite-index in H_ψ, the H_ψ-orbit of $\mathbb{P}(W^\perp)$ is finite. This contradicts Corollary 3.5. □

Proposition 5.4 (Full Prym support). *Let $x \in V_\psi \otimes U$ be non-zero. If the K-orbit $[x]$ has finite H_ψ-orbit, then*

$$S(x) = V_\psi.$$

Proof. Since $[x]$ has finite H_ψ-orbit, there is a finite-index subgroup $H_0 \leq H_\psi$ stabilizing $[x]$. Thus for every $h \in H_0$ there exists $k_h \in K$ such that $hx = k_h x$. By Lemma 5.2,

$$hS(x) = S(hx) = S(k_h x) = S(x).$$

Thus $S(x)$ is fixed by H_0. Since $x \neq 0$, $S(x) \neq 0$. By Lemma 5.3, it cannot be a proper non-zero subspace. Hence $S(x) = V_\psi$. □

Corollary 5.5 (Injectivity of the contraction map). *Under the hypotheses of Proposition 5.4, the contraction map*

$$T_x : V_\psi^\vee \longrightarrow U = \mathrm{Hom}(N, M)$$

associated with x is injective.

Proof. The dual map $T_x^\vee : U^\vee \to V_\psi$ has image exactly $S(x)$. Thus $S(x) = V_\psi$ is equivalent to surjectivity of $T_x^\vee$, which is equivalent to injectivity of T_x. □

6. Rank strata and constant rank

Assume T_x is injective and set

$$L := T_x(V_\psi^\vee) \subset U = \mathrm{Hom}(N, M).$$

Then

$$\dim L = \dim V_\psi = 2g - 2.$$

Definition 6.1 (Pulled-back determinantal strata). For $0 \leq r \leq \min(m, n)$, let

$$D_r = \{A \in \mathrm{Hom}(N, M) : \mathrm{rank}\, A \leq r\}$$

be the usual determinantal variety. Define

$$Z_r(x) = \{[\ell] \in \mathbb{P}(V_\psi^\vee) : \mathrm{rank}\, T_x(\ell) \leq r\}.$$

This is a closed subvariety of $\mathbb{P}(V_\psi^\vee)$ because D_r is cut out by the $(r+1) \times (r+1)$ minors.

Lemma 6.2 (Rank strata are K-orbit invariants). *For every $k \in K$ and every r,*

$$Z_r(kx) = Z_r(x).$$

Proof. Write $k = (A, B) \in \mathrm{GL}(M) \times \mathrm{GL}(N)$. The map attached to kx is

$$T_{kx}(\ell) = A T_x(\ell) B^{-1}.$$

Left and right multiplication by invertible matrices preserve rank. □

Lemma 6.3 (Finite orbit of rank strata). *If the K-orbit $[x]$ has finite H_ψ-orbit, then each $Z_r(x)$ has finite H_ψ-orbit as a closed subvariety of $\mathbb{P}(V_\psi^\vee)$.*

Proof. Using the convention fixed in Appendix A, if h acts on V_ψ on the left, then

$$T_{hx}(\ell) = T_x(h^{-1}\ell)$$

for the contragredient action on $V_\psi^\vee$. Hence

$$Z_r(hx) = hZ_r(x).$$

By Lemma 6.2, replacing hx by a K-equivalent tensor does not change the rank stratum. Since the H_ψ-orbit of $[x]$ is finite, only finitely many closed subvarieties Z_r can occur. □

Proposition 6.4 (Finite-orbit rank strata force constant rank)**.** *Let $x \in V_\psi \otimes U$ have finite H_ψ-orbit in $(V_\psi \otimes U)/K$, and suppose*

$$T_x : V_\psi^\vee \longrightarrow U$$

is injective. Then every non-zero element of

$$L = T_x(V_\psi^\vee)$$

has the same rank.

Proof. By Lemma 6.3, each $Z_r(x)$ has finite H_ψ-orbit. By Corollary 3.5, each $Z_r(x)$ is either empty or all of $\mathbb{P}(V_\psi^\vee)$. Since T_x is injective, $T_x(\ell) \neq 0$ whenever $0 \neq \ell \in V_\psi^\vee$.

If the non-zero elements of L do not all have the same rank, let

$$r_0 = \min\{\operatorname{rank} A : 0 \neq A \in L\}.$$

Then $Z_{r_0}(x)$ is non-empty, but it is not all of $\mathbb{P}(V_\psi^\vee)$ because some non-zero element of L has rank larger than r_0. This contradicts Corollary 3.5. □

7. Westwick bound and proof of optimality

External Theorem 7.1 (Westwick)**.** *Let $2 \le r \le m \le n$. If*

$$L \subset \operatorname{Hom}(\mathbb{C}^n, \mathbb{C}^m)$$

is a complex linear subspace such that every non-zero element of L has rank r, then

$$\dim L \le m + n - 2r + 1.$$

This is the form of Westwick's fixed-rank theorem needed below. It is also quoted in Rubei's survey on affine subspaces of matrices with rank in a range.

Lemma 7.2 (Weak constant-rank bound)**.** *Let*

$$L \subset \operatorname{Hom}(\mathbb{C}^n, \mathbb{C}^m)$$

be a complex linear subspace such that every non-zero element of L has the same rank $r \ge 1$. Then

$$\dim L \le m + n - 1.$$

Proof. If $r \geq 2$, transpose all matrices if necessary so that the smaller dimension is the target dimension. External Theorem 7.1 gives

$$\dim L \leq m + n - 2r + 1 \leq m + n - 1.$$

It remains to treat $r = 1$. Choose a non-zero element $A = a \otimes \alpha \in L$ with $a \in \mathbb{C}^m$ and $\alpha \in (\mathbb{C}^n)^\vee$. If $B = b \otimes \beta \in L$, then every linear combination $A + tB$ has rank at most one. Therefore

$$(a \wedge b) \otimes (\alpha \wedge \beta) = 0.$$

Thus for every $B = b \otimes \beta \in L$, either b is proportional to a, or β is proportional to α.

One of these alternatives holds for all $B \in L$. Indeed, if there were $B = b \otimes \beta$ and $C = c \otimes \gamma$ with b not proportional to a and γ not proportional to α, then the previous paragraph forces β proportional to α and c proportional to a. Then $B + C$ has two independent image directions and two independent covectors, hence rank two, a contradiction. Therefore all elements of L have a common image line, or all elements factor through a common one-dimensional quotient. In the first case $\dim L \leq n$; in the second case $\dim L \leq m$. Hence

$$\dim L \leq \max(m, n) \leq m + n - 1.$$

□

Proof of Theorem 1.2. Let $x \in V_\psi \otimes U$ be the extension class of ρ. By Lemma 4.1, the infinite-image assumption gives $x \neq 0$. By Lemma 4.3, the ordinary finite mapping-class-group orbit gives a finite H_ψ-orbit of the K-orbit $[x]$. Therefore Proposition 5.4 gives $S(x) = V_\psi$, and Corollary 5.5 gives an injection

$$T_x : V_\psi^\vee \hookrightarrow \mathrm{Hom}(N, M).$$

Set

$$L = T_x(V_\psi^\vee).$$

Then

$$\dim L = \dim V_\psi = 2g - 2.$$

By Proposition 6.4, every non-zero element of L has the same rank. By Lemma 7.2,

$$2g - 2 = \dim L \leq \dim M + \dim N - 1.$$

Hence

$$\dim M + \dim N \geq 2g - 1.$$

□

8. Consequences and the genus-three case

For every $g \geq 2$ and every non-trivial finite character ψ, Theorem 1.1 produces a rank $2g-1$ non-semisimple representation with infinite image and finite ordinary mapping-class-group orbit. Theorem 1.2 says that this rank cannot be improved inside the scalar two-block triangular Prym category.

When $g = 3$, one has

$$\dim V_\psi = 4, \qquad 2g - 1 = 5.$$

Thus rank 5 examples exist. A rank 4 scalar two-block compression would have $m + n \leq 4$. Full support would give an injection

$$V_\psi^\vee \hookrightarrow \mathrm{Hom}(N, M),$$

so $4 \leq mn$. The only positive integers with $m+n \leq 4$ and $mn \geq 4$ are $m = n = 2$. Hence a rank 4 scalar two-block compression would identify $V_\psi^\vee$ with a four-dimensional space of 2×2 matrices. Pulling back the determinant would give a finite-orbit projective invariant, contradicting the Prym monodromy consequence above. The general proof replaces this determinant line by all determinantal rank strata and works for every genus and every pair (m, n).

Consequently, in genus three, any rank < 5 finite-orbit infinite-image example, if it exists, must come from outside the scalar two-block Prym framework: for instance from non-scalar finite-image semisimple blocks, longer filtrations, or another non-Prym mechanism. No such global exclusion is proved here.

9. Logical dependency checklist

The proof can be audited through the following dependency chain.

Step	Content
1	For non-trivial finite ψ, $\dim H^1(\Sigma_g;\mathbb{C}_\psi) = 2g-2$.
2	The identity tensor in $V_\psi \otimes V_\psi^\vee$ gives a rank $2g-1$ triangular representation.
3	Its cocycle is non-zero on $\ker\psi$, hence the image contains a non-trivial unipotent element and is infinite.
4	The finite character orbit and the universal nature of the identity tensor give finite ordinary mapping-class-group orbit.
5	Looijenga's cyclic Prym theorem, after the translations in Section 3.1 and Appendix B, gives the single-character projective Zariski closure: PSp if $\psi^2 = 1$, and PSL otherwise.
6	Finite-orbit closed subvarieties of $\mathbb{P}(V_\psi^\vee)$ are empty or all.
7	Ordinary finite orbit of a scalar two-block extension gives finite H_ψ-orbit of $[x] \in (V_\psi \otimes U)/K$.
8	Infinite image gives $x \neq 0$.
9	Finite orbit and Prym monodromy force $S(x) = V_\psi$.
10	Full support is equivalent to injectivity of $T_x : V_\psi^\vee \to \mathrm{Hom}(N, M)$.
11	Determinantal rank strata are closed K-orbit invariants.
12	Prym monodromy forces $T_x(V_\psi^\vee)$ to have constant non-zero rank.
13	Westwick's constant-rank bound gives $\dim T_x(V_\psi^\vee) \leq \dim M + \dim N - 1$.
14	Since $\dim T_x(V_\psi^\vee) = 2g-2$, one gets $\dim M + \dim N \geq 2g-1$.

The only mapping-class-group input is Looijenga's cyclic image theorem in External Theorem 3.1. Lemma 3.3 also uses the standard arithmetic-group Zariski-density/base-change fact for the corresponding K^+-algebraic special unitary group. The other external linear-algebra input is Westwick's theorem, External Theorem 7.1. All other steps are cohomological bookkeeping, projective orbit arguments, or elementary linear algebra.

Appendix A. Action conventions for the contraction map

Let $x \in V \otimes U$, and let

$$T_x : V^\vee \longrightarrow U$$

be the contraction map

$$T_x(\ell) = (\ell \otimes \mathrm{id}_U)(x).$$

If a group element h acts on V on the left, then it acts on $V^\vee$ by the contragredient rule

$$(h\ell)(v) = \ell(h^{-1}v).$$

Writing $x = \sum_i v_i \otimes u_i$, one obtains

$$T_{hx}(\ell) = \sum_i \ell(hv_i)u_i = T_x(h^{-1}\ell).$$

Consequently $Z_r(hx) = hZ_r(x)$ for the rank loci used in Lemma 6.3. If one adopts the opposite left/right convention, all occurrences of h in this formula are replaced by h^{-1}; the orbit and finite-orbit arguments are unchanged.

Appendix B. Verification of the Looijenga input

The single-character consequence in Proposition 3.4 is obtained from Looijenga's cyclic theorem through the following finite list of translations. The notation S_C, $S_C^{\#}$, H_C, R_C, and $U^{\#}(H_C)$ is Looijenga's notation from Section 2 of [1], except that $S_C^{\#}$ abbreviates his lifted group $\mathcal{S}^{\#}_{;C}$.

(A1) **Cyclic quotient.** Given ψ, choose the finite cyclic quotient $q : H_1(\Sigma_g;\mathbb{Z}) \twoheadrightarrow C$ such that

$$q^*(\widehat{C}) = \langle\psi\rangle \subset \mathrm{Hom}(H_1(\Sigma_g;\mathbb{Z}), \mathbb{C}^\times).$$

Looijenga's subgroup S_C is the subgroup of mapping classes inducing the identity on C, equivalently fixing $\widehat{C}$ pointwise under the dual action. Since $q^*(\widehat{C})$ is generated by ψ, fixing ψ is equivalent to fixing every power of ψ, hence to fixing $q^*(\widehat{C})$ pointwise. Thus S_C is the group denoted here by H_ψ. This is stronger than merely preserving the kernel of q.

(A2) **Lifted group.** Looijenga's Prym representation is naturally defined for a lifted group $S_C^{\#}$ fitting into the central extension

$$1 \longrightarrow C \longrightarrow S_C^{\#} \longrightarrow S_C \longrightarrow 1.$$

The kernel is the deck group. On the fixed ψ-eigenspace, a deck transformation $c \in C$ acts by the scalar $\psi(c)$ or by $\psi(c)^{-1}$, depending on left/right module convention. In either case the action is scalar, hence it maps trivially to $\mathrm{PGL}(E_\psi)$. Therefore $S_C^{\#}$ and $S_C = H_\psi$ have the same connected projective Zariski closure on $\mathbb{P}(E_\psi)$.

(A3) **Prym module.** Looijenga's Proposition 2.2 gives that H_C is a free R_C-module of rank $2(g-1)$ with its skew-Hermitian intersection pairing. After scalar extension $R_C \to \mathbb{C}$ determined by ψ, the block

$$E_\psi = H_C \otimes_{R_C,\psi} \mathbb{C}$$

identifies, with our convention, with the twisted homology block

$$H_1(\Sigma_g; \mathbb{C}_{\psi^{-1}}),$$

and hence

$$E_\psi^\vee \cong H^1(\Sigma_g; \mathbb{C}_\psi) = V_\psi.$$

Under the opposite convention, replace ψ by ψ^{-1}; this does not affect the final projective closure conclusion.

(A4) **Image theorem.** Looijenga's Theorem 2.4 identifies the image of $S_C^{\#}$ on H_C with $U^{\#}(H_C)$ in the cyclic case. This is the only deep Prym monodromy input used in the proof.

(A5) **Classical group after scalar extension.** Write $K = K_C$ and $K^+ = K_C^0$. Under Looijenga's identification of the image with the arithmetic group $U^\#(H_C)$, the connected complex Zariski closure is obtained by base-changing the corresponding K^+-algebraic special unitary group. The finite determinant and root-of-unity conditions in $U^\#(H_C)$ only add finite cosets and do not affect the identity component of the complex projective Zariski closure.

If $\psi^2 = 1$, then since $\psi \neq 1$, the character has order two. The block E_ψ is self-dual and Looijenga's skew-Hermitian form becomes a non-degenerate alternating complex bilinear form on E_ψ. The connected complex group on the block is therefore symplectic, not orthogonal, and the connected projective group is $\mathrm{PSp}(E_\psi)$.

If $\psi^2 \neq 1$, then after base change through the real embedding of K^+ determined by $\psi|_{K^+}$,

$$H_C \otimes_{K^+} \mathbb{C} = E_\psi \oplus E_{\psi^{-1}}.$$

The skew-Hermitian form pairs E_ψ perfectly with $E_{\psi^{-1}}$, and each summand is isotropic. The complexified special unitary group acts as

$$A \oplus (A^{-1})^\vee, \qquad A \in \mathrm{SL}(E_\psi).$$

Thus the connected projective group on E_ψ is $\mathrm{PSL}(E_\psi)$.

(A6) **Finite-index subgroups.** If $H_0 \leq H_\psi$ is a finite-index subgroup, the image of H_ψ is a finite union of translates of the image of H_0. Their Zariski closures therefore have the same identity component.

No statement about simultaneous independence of several eigenspaces is invoked, and no theorem about finite orbits in a quotient is used.

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[Affiliation]
Email address: [Email]